

Satiety: have we neglected dietary non-nutrients?

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Satiety, which is the inhibition of eating following the end of a meal, is influenced by a number of food characteristics, including compositional and structural factors. An increased understanding of these factors and the mechanisms whereby they exert their effects on satiety may offer a food-based approach to weight management. Water and gas, which are often neglected in nutrition, are major components of many foods and contribute to volume, and to sensory and other characteristics. A review of previous short-term studies that evaluated the effects of water or gas in foods on satiety showed that while satiety was generally increased, effects on subsequent intakes were not always apparent. These studies were diverse in terms of design, timings and food matrices, which precludes definitive conclusions. However, the results indicate that solids may be more effective at increasing satiety than liquids, but gas may be as effective as water. Although increased gastric distension may be the main mechanism underlying these effects, pre-ingestive and ingestive impacts on cognitive, anticipatory and sensory responses also appear to be involved. Furthermore, there is limited evidence that water on its own may be effective at increasing satiety and decreasing intakes when drunk before, but not with, a meal. Longer-term extrapolation suggests that increasing food volumes with water or gas may offer weight-management strategies. However, from a practical viewpoint, the effects of water and gas on satiety may be best exploited by using these non-nutrients to manipulate perceived portion sizes, without increasing energy contents.